

HAL

BOOK II — ARCHITECTURE SPECIFICATION

v0.1 • Chapter 1 complete

Book II Charter

Book I defines what must remain true. Book II defines the technical architecture that fulfills those constitutional requirements. Book II is governed by Book I; no architectural decision may weaken or contradict a constitutional rule.

Authoring Workflow

- Architect draft
- Constitutional traceability audit
- Internal engineering review
- Revision of internally resolvable findings
- Owner review only for constitutional, human-value, irreversible-risk, or authority choices

Planned Book Structure

Part	Planned chapters
I — Architectural Foundations	Overall system architecture; runtime; service and event model; data architecture; identity, trust, and capability foundations.
II — Core Services	Kernel, policy, identity, authority, intent, knowledge, learning, attention, judgment, verification, audit, recovery, treaty, presence, cluster.
III — Infrastructure	Storage, networking, messaging, scheduling, caching, security, secrets, replication, observability, deployment, updates, and recovery.
IV — Interfaces & Reference	Presence interfaces, APIs, automation, developer interfaces, schemas, events, protocol references, and conformance.

Chapter 1 — Overall System Architecture

Field	Value
Status	Architect draft complete; constitutional and engineering review complete
Version	0.1
Scope	System topology, authority boundaries, major planes, data ownership, and end-to-end flow
Constitutional basis	Decisions 1–58, with primary implementation responsibility for 1–7, 10–16, 24–25, 29, 32, 35–42, 45, 47, 49–51
Owner review	None required at this stage

1. Purpose

This chapter defines the whole HAL system before later chapters specify individual services. It establishes which components are authoritative, how data and actions cross boundaries, and how the system preserves constitutional identity while executing work through replaceable models, services, nodes, and external partners.

2. Architectural Thesis

HAL is a constitutionally governed control system. The constitutional kernel makes and records authoritative decisions; governed services reason, plan, store, observe, and coordinate; capability providers perform bounded work; Presences expose the system to people and the physical world. No provider, node, or interface gains authority merely by participating.

3. System Planes

Plane	Responsibilities	Authority posture
Constitutional plane	Owner identity, policy, authority, authorization ceremony, audit integrity, constitutional state.	Authoritative; only governed mutation paths.
Cognitive-control plane	Intent, planning, evidence synthesis, trust, attention, judgment, scheduling, verification.	Makes or recommends decisions under policy.
State and knowledge plane	Experience Ledger, knowledge graph, decision records, configuration, transactions, outcomes.	Each durable domain has one authoritative owner.
Execution plane	Execution Orchestrator, capability contracts, providers, node runtimes, adapters, transactions.	Bounded execution; cannot alter constitutional state directly.

Plane	Responsibilities	Authority posture
Presence and federation plane	Desktop, phone, voice, watch, robot, APIs, external trust domains, Treaties.	Contextual interfaces; cross-domain exchange passes the Constitutional Firewall.

4. Reference Topology

Owner & Presences

↓ authenticated, authorized requests

HAL Core — Constitutional Kernel + Governed Services

- └ Constitutional plane: identity • policy • authority • audit
 - └ Cognitive-control plane: intent • planning • evidence • trust • judgment • scheduling
 - └ State plane: ledger • knowledge • transactions • configuration • observability
 - └ Execution Orchestrator
- ↓ capability contract + delegated authority

HAL Node Runtimes & Governed Providers

↓ bounded results, evidence, health, and transaction updates

External Trust Domains / Physical & Digital Environment

5. Constitutional Kernel

The kernel is deliberately small. It contains only what must remain trustworthy when every optional service fails: root identity, authorization, policy issuance and evaluation authority, protected audit rules, transaction commit coordination, cluster membership authority, and recovery control. It exposes narrow typed contracts. All other services are replaceable and interact with the kernel by commands, queries, and durable events.

6. Authoritative State Ownership

State domain	Authoritative owner	Derived / replaceable forms
Identity, delegation, authentication	Identity & Authority Service	Session caches, UI claims, presence assertions
Policy, exceptions, approvals	Policy System	Signed evaluation bundles, local evaluators
Experience and audit	Experience Ledger / Audit Service	Indexes, summaries, forensic projections
Knowledge and patterns	Knowledge Service	Embeddings, caches, search indexes
Intent, plan, transaction, outcome	Intent Manager / Planner / Transaction Coordinator	Dashboards, work queues, provider-specific task representations
Configuration and secrets references	Configuration Plane / Secrets Service	Node-local verified bundles, short-lived credentials
Node and provider observations	Node Registry / Provider Registry	Health views, benchmarks, scheduling projections

7. Request-to-Outcome Flow

- A Presence submits a structured request to HAL Core. The request carries identity, context, intent, priority, evidence, classification, and correlation metadata.
- Identity, policy, and trust checks establish whether the request can be understood, planned, or acted upon.
- The Intent and Planning services derive an approved plan. The Decision Engine records alternatives, evidence, uncertainty, and policy constraints.
- The Execution Orchestrator binds capability contracts to current providers and nodes, opens a transaction, reserves resources, and runs only within delegated authority.
- Providers return results, evidence, health, and execution facts. They never directly mutate authoritative state.
- Verification evaluates the result. The transaction completes, compensates, pauses, recovers, or escalates. Experience, audit, knowledge, and outcome records are updated through their authoritative owners.

8. Trust Boundaries

Boundary	Required control
Presence ↔ HAL Core	Continuous authentication, authorization, disclosure appropriate to Presence privacy and confidence.
Kernel ↔ governed service	Typed contract, least privilege, correlation, auditable policy decision, no direct database ownership violation.
Core ↔ node runtime	Mutual authentication, encrypted transport, signed configuration, health evidence, delegated execution envelope.
Provider ↔ adapter	Capability-scoped credential, provider fitness evidence, timeout/retry semantics, output verification.
HAL ↔ external trust domain	Treaty, Constitutional Firewall, provenance validation, data minimization, revocation and audit.

9. Operating Modes & Failure Posture

HAL exposes Normal, Degraded, Restricted, Safe Mode, Recovering, Maintenance, Emergency, and Offline modes. A failure may reduce availability or capability but cannot silently change policy, identity, authority, authentication requirements, trust thresholds, or constitutional rules. Critical state is preserved through authoritative ownership, append-only journals, durable transaction checkpoints, independent backups, and recovery-first boot.

10. Deployment Baseline

The initial deployment uses a Mac mini as Primary Constitutional Host and HAL Core control plane; GX10 systems run HAL Node Runtimes for governed model and specialist execution; a NAS holds a warm, encrypted, restore-tested recovery replica plus independent backup. This is a deployment topology, not a permanent identity requirement. HAL remains portable as a continuity of constitution, governed state, and authority.

11. Interfaces

Interface class	Primary semantic
Command	Requests one authoritative state change; caller receives

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	acceptance, rejection, or required approval.
Query	Reads declared consistency class from an authoritative owner or controlled projection.
Event	Immutable fact with identity, schema, correlation, causation, authorization, provenance, and integrity metadata.
Capability invocation	Bounded request to fulfill a capability contract under a transaction and delegation envelope.
Treaty exchange	Cross-domain message governed by an active Treaty and the Constitutional Firewall.

12. Guarantees

- No external model, provider, node, service, or Presence is a constitutional authority by default.
- Every durable authoritative state domain has one owner, versioned mutation, append-only history, and auditable recovery path.
- Every meaningful action has a transaction, policy context, authorization context, evidence, and explainable outcome.
- All cross-process, device, or trust-boundary communication is authenticated, integrity-protected, and encrypted; sensitive exchanges are end-to-end protected.
- Degradation is disclosed. System rules do not silently change because a dependency fails.
- The architecture remains implementation-neutral: models, adapters, nodes, and non-kernel services are replaceable without changing HAL identity.

13. Constitutional Traceability Audit

Constitutional decisions	Chapter 1 implementation coverage
1–7	Core boundary, modular services, CQE contracts, event journal, state ownership, immutable identity, Self Model.
8–16	Engagement tiers, decision framework, capabilities, knowledge, planning, provider registry, graph separation, orchestration.
17–25	Nodes, resource constraints, lifecycle, work model, scheduling, recovery, communication, consistency, policy.
26–35	Trust, authentication, introspection, distributed execution, memory, intent, communication, learning, reasoning, transactions.
36–45	Capability stack, kernel boundary, supervision,

Constitutional decisions	Chapter 1 implementation coverage
	configuration, observability, resources, durability, updates, time, Presences.
46–58	Initiative, coordination, ownership, sovereignty, verification, self-description, environment, wisdom, attention, judgment, restraint, outcomes, evolution.

14. Internal Engineering Review

Check	Result
Constitutional conflicts	None identified. Book I remains the governing authority.
Authority ambiguity	Resolved through single-owner durable state and kernel-only constitutional mutation.
Failure posture	Explicit modes, safe degradation, recovery journals, and independent backup topology included.
Premature technology coupling	Avoided. Specific transports, databases, and frameworks are deferred to later chapters.
Owner review required	None. This chapter chooses implementation structure without altering constitutional philosophy or Owner authority.

15. Completion Status

Chapter 1 is complete as an architectural foundation. Subsequent chapters will refine the runtime, services, data models, interfaces, and deployment mechanisms named here without changing the authority model established in this chapter.